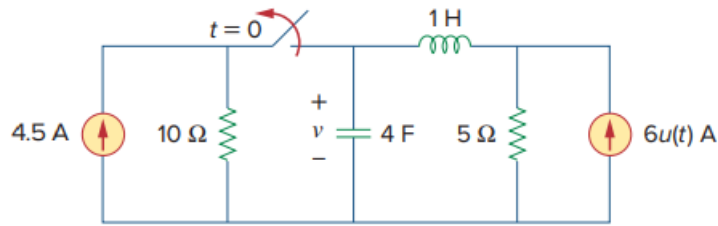


Problem

Find $v(t)$ for $t > 0$ in the circuit of Fig. 8.81.

Figure 8.81:



Step-by-step solution

Step 1 of 6

The circuit is in steady state at $t = 0^-$.

Draw the equivalent circuit at $t = 0^-$.

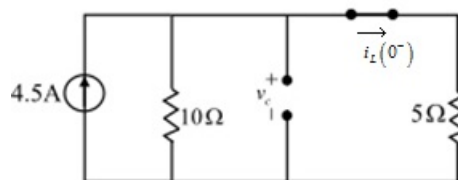


Figure 1

Step 2 of 6

Determine the initial conditions of the circuit.

Use current division principle to determine the inductor current at $t = 0^-$.

$$\begin{aligned} i_L(0^-) &= 4.5 \times \left(\frac{10}{5+10} \right) \\ &= 4.5 \times \left(\frac{2}{3} \right) \\ &= 3 \text{ A} \end{aligned}$$

Determine the voltage across the capacitor at $t = 0^-$.

$$\begin{aligned} v_C(0^-) &= 5i_L(0^-) \\ &= 5(3) \\ &= 15 \text{ V} \end{aligned}$$

Draw the equivalent circuit for $t > 0$ as shown in Figure 2.

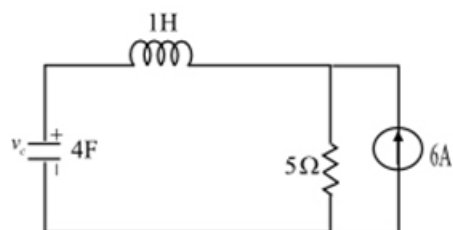


Figure 2

Step 3 of 6

Use source transformation theorem to replace the 6 A source with its equivalent voltage source.

$$\begin{aligned} V &= 5 \times 6 \\ &= 30 \text{ V} \end{aligned}$$

Draw the equivalent circuit as shown in Figure 3.

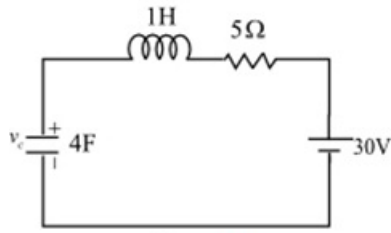


Figure 3

Step 4 of 6

The expression for the inductor current is,

$$i_L(t) = -C \frac{dv_C(t)}{dt}$$

Substitute 0 for t .

$$i_L(0) = -C \frac{dv_C(0)}{dt}$$

$$\frac{dv_C(0)}{dt} = -\frac{i_L(0)}{C}$$

Substitute 3 A for $i_L(0)$ and 4 F for C .

$$\begin{aligned} \frac{dv_C(0)}{dt} &= -\frac{3}{4} \\ &= -0.75 \end{aligned}$$

Apply Kirchhoff's voltage law in the circuit shown in Figure 3.

$$-v_C + (1) \frac{di_L}{dt} + 5i_L + 30 = 0$$

$$-\frac{di_L}{dt} - 5i_L + v_C = 30$$

Substitute $-C \frac{dv_C}{dt}$ for i_L .

$$-\frac{d}{dt} \left(-C \frac{dv_C}{dt} \right) - 5 \left(-C \frac{dv_C}{dt} \right) + v_C = 30$$

$$C \frac{d^2 v_C}{dt^2} + 5C \frac{dv_C}{dt} + v_C = 30$$

Substitute 4 F for C .

$$4 \frac{d^2 v_C}{dt^2} + 20 \frac{dv_C}{dt} + v_C = 30 \quad \dots (1)$$

Step 5 of 6

Determine the roots of the characteristic equation for equation 1.

$$4D^2 + 20D + 1 = 0$$

$$s_1 = -0.0505, s_2 = -4.95$$

The type of the output response is overdamped, since, the roots of the characteristic equation are real and distinct.

Refer to Equation 8.44a in the text book and write the expression of the overdamped response.

$$v_C(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Substitute 30 V for V_s , -0.0505 for s_1 and -4.95 for s_2 .

$$v_C(t) = 30 + A_1 e^{-0.0505t} + A_2 e^{-4.95t} \dots\dots (2)$$

Determine the coefficients A_1 and A_2 using the initial conditions.

Substitute 0 for t and 15 V for $v_C(0)$.

$$15 = 30 + A_1 e^{-0} + A_2 e^{-0}$$

$$A_1 + A_2 = -15 \dots\dots (3)$$

Step 6 of 6

Differentiate equation 2 with respect to t .

$$\frac{dv_C}{dt} = -0.0505 A_1 e^{-0.0505t} - 4.95 A_2 e^{-4.95t}$$

Substitute 0 for t and -0.75 A/F for $\frac{dv_C(0)}{dt}$.

$$-0.75 = -0.0505 A_1 - 4.95 A_2 \dots\dots (4)$$

Solve equations 3 and 4.

$$A_1 = -15.308$$

$$A_2 = 0.308$$

The expression of the voltage across the capacitor is,

$$\begin{aligned} v_C(t) &= v(t) \\ &= 30 - 15.308 e^{-0.0505t} + 0.308 e^{-4.95t} \end{aligned}$$

Thus, the expression of $v(t)$ is $\boxed{30 - 15.308 e^{-0.0505t} + 0.308 e^{-4.95t} \text{ V}}$.